MIL-P-36556A
10 July 1976
SUPERSEDING
MIL-P-36556
18 March 1968

MILITARY SPECIFICATION

PIN, TOOTH DIE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a universal dowel pin suitable for use in fabrication of removable dies in dental prosthetic laboratory procedures.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-B-626	- Brass, Leaded and Nonleaded, Rod, Shaped, Forgings, and Flat Products with Finished Edges (Bar and Strip).
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

MILITARY

MIL-P-116	- Methods of preservation-packaging.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, Attn: Directorate of Medical Materiel, CODE ATT, 2800 South 20th St., Phila. PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6520

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

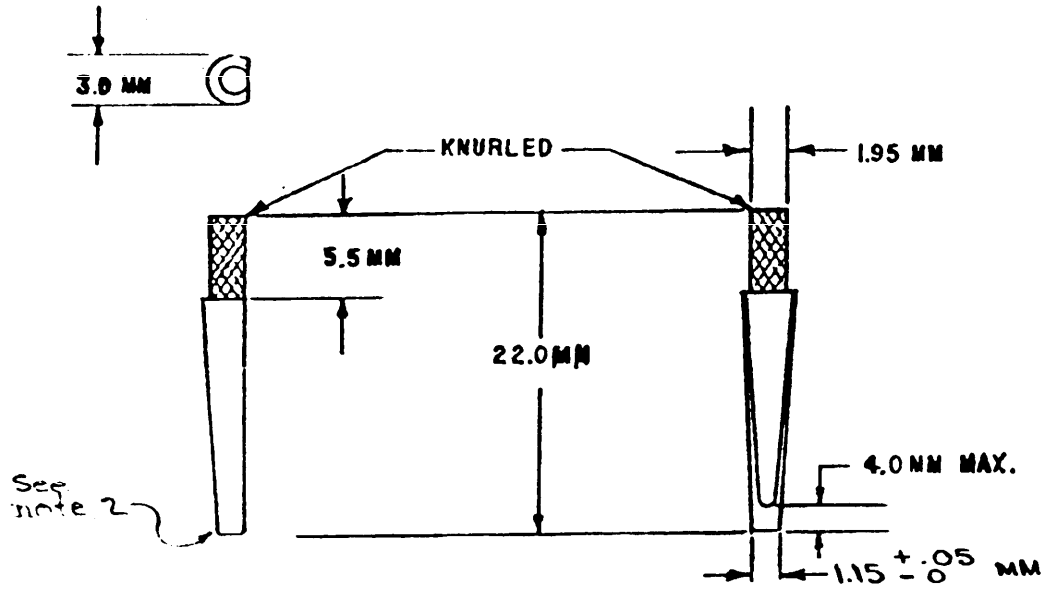
3.1 **Material**. **The** material shall conform to composition 11, half hard brass, of QQ-B-626 for composition and hardness.

3.2 Design and dimensions. The tooth die pins shall be universal dowel pins suitable for use in fabrication of removable dies in dental prosthetic laboratory procedures. The tooth die pins shall be of one-piece construction. The pins shall have a 35-teeth-per-inch-diamond knurl on their retention heads to insure a secure relationship between pin and die when bonded with a suitable material. The remaining portions of the pin shall be free from irregularities to permit ease of removal and reinsertion into a working cast. The pins shall have a round tapering shaft, one side of which shall be furnished with a flat surface to serve as a positive-seating key when used as intended. The tooth die pins shall conform to either style of figure 1 for design and dimensions. The dimensions of one style shall not be interchanged with the other style.

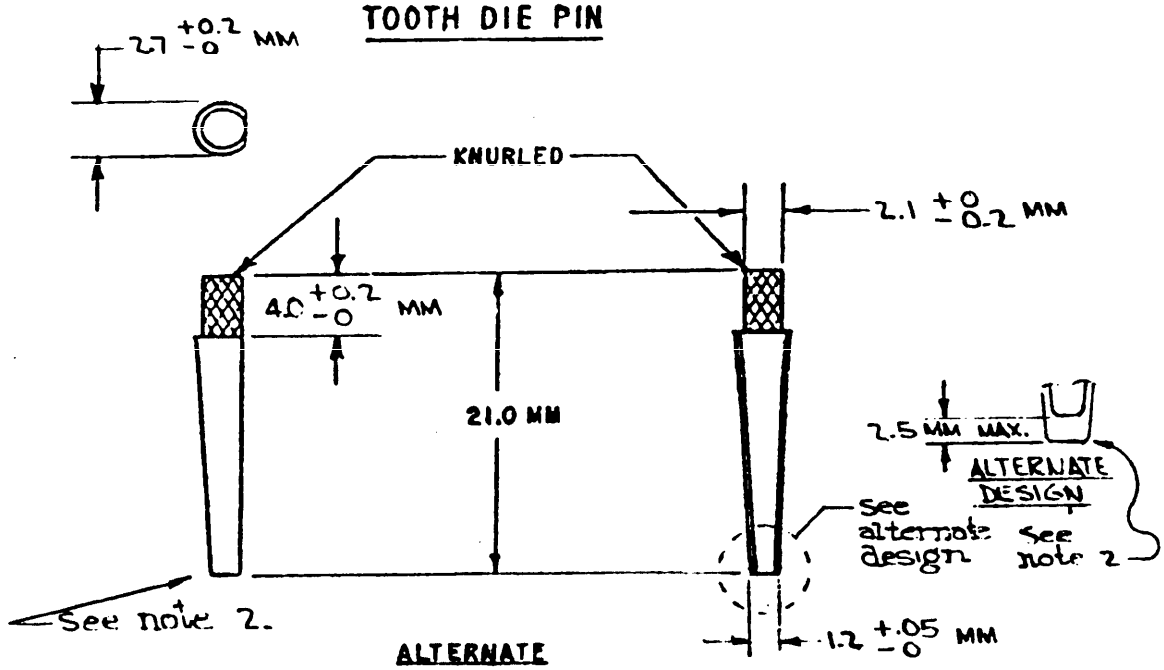
3.2.1 Figures. The figures forming a part of this specification are not production drawings. These figures establish dimensional limitations and control design and configuration. The supplier shall be responsible for preparing his own shop drawings. The supplier shall select dimensions and tolerances within the limitations prescribed on the figures to provide a product with proper design, configuration and function. No deviations from the dimensional limitations prescribed on the figures are permissible without prior approval of the contracting officer.

3.3 Finish. The die pin, other than the knurled section, shall have a satin or better finish.

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TOOTH DIE PIN



ALTERNATE TOOTH DIE PIN

NOT TO SCALE:

Notes:

1. UNLESS OTHERWISE SPECIFIED TOLERANCE ON DIMENSIONS IS ±5%
2. No burrs, break sharp edges.

FIGURE 1

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3.4 Workmanship. Workmanship shall be first class throughout. The finished tooth die pins shall be free from defects which detract from their appearance, or may impair their serviceability, proper functioning, or intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the brass, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

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4.2.1 Examination. The die pins shall be examined to determine compliance with requirements contained in this specification. Examination shall be conducted in accordance with the following classification of defects. Examination shall include, but shall not be limited to, the defects listed in table I.

TABLE I. Classification of defects.

Categories	Defects
Critical:	None defined.
Major A:	
101	Serrations of knurled section not as specified.
102	Style and design of pin not as specified.
103	Body of pin has surface irregularities.
104	Burrs and slivers present.
Major B:	
251	Dimensions not as specified.
Minor:	
201	Finish not as specified.
202	Pin covered with oil, dirt and other foreign matter.
203	Pin grossly discolored.

The inspection levels and acceptable quality levels (AQLs) shall be as listed in table II.

TABLE II. Examination criteria.

Type defect	Inspection level	AQL (Percent defective)
Major A	II	1.0
Major B	S-2	4.0
Minor	II	2.5

4.2.2 Testing. Pins shall be tested as specified in table III. Inspection levels and acceptable quality levels (AQLs) shall be as listed in table III. The unit of product for test purposes shall be one tooth die pin.

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TABLE III. Tests.

Characteristic	Requirement	Test procedure	Inspection level	AQL
Hardness	3.1	QQ-B-626	S-2	1.5
Composition	3.1	QQ-B-626	S-1*	**

* A sufficient quantity from each lot of metal used in the construction of the die pins shall be selected for composition analysis in lieu of sample die pins, provided lot identity has been maintained and can be demonstrated to the Government.

**Acceptance number shall be zero for all sample sizes.

4.3 Tests. Tests shall be conducted to determine compliance with the requirements of this specification (see 4.2.2).

4.4 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.1).

5.1.1 Unit of issue. One package containing one hundred pins, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Cleaning. Pins shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.2.2 Drying. Pins shall be dried in accordance with MIL-P-116.

5.1.2.3 Preservative application. A preservative shall not be applied to the pins.

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5.1.2.4 Unit packaging. One hundred pins shall be packaged in a plastic container having a screw cap closure.

5.1.2.5 Intermediate package. Twelve unit packages shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the box specification.

5.1.3 Level C. One hundred pins shall be clean, dry and packaged in a manner that will afford adequate protection against corrosion or physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. Twenty four intermediate packages (288 units) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Twenty-four intermediate packages (288 units) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged pins shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.3 Marking.

5.3.1 Interior packages. Each unit and intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.3 National Stock Number (NSN) coverage. This specification covers the following item appearing in the Federal Supply Catalog:

NSN	Item Identification
6520-00-930-8046	PIN, TOOTH DIE, 100s

MILITARY CUSTODIANS:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA-DM

Project Number: 6520-1422

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-P-36556A - PIN, TOOTH DIE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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